

Executive summary

This report focuses on developing a predictive model to identify the likelihood of customers to purchase ABC bank's variable annuity product for targeted marketing campaigns. The dataset used consists of 32,264 banking customers, 47 input variables, and a binary target variable. The objective is to optimise resource allocation and increase campaign success by accurately identifying potential customers. The report includes data preprocessing, selective predictive modelling techniques, evaluation metrics, and feature importance is interpreted to identify the champion model using SAS Viya. The champion model with the highest predictive accuracy based on the lowest misclassification rate is determined and integrated into the decision-making process. The findings will provide potential implications for ABC bank's marketing campaign, enabling effective targeting and improved business outcomes.

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1. Introduction

Business problem: ABC bank conducted a target marketing campaign to promote a variable annuity (insurance product). The purpose of this campaign was to identify segments of customers who are likely to respond positively to the campaign. To achieve this goal, a dataset was created with information on 32,264 banking customers, 47 input variables and a binary target variable.

Objective: To accurately identify customers who are more likely to purchase the variable annuity product based on their demographic information and product usage patterns.

This report builds upon the previously submitted proposal which outlined the background and context of the business problem, emphasising details such as the scope of work and timeline of report. I will provide a comprehensive analysis of the predictive modelling process focusing on the variable annuity product. I will provide an overview of the dataset used for the analysis, including its source and composition. I will ensure the quality of the data by outlining the data preprocessing and explain the selected predictive modelling techniques suited for this model. Based on the evaluation metrics and performance analysis, I will identify the champion model with the lowest misclassification rate and interpret the feature importance. Finally, I will summarise the findings and implications of the predictive modelling analysis for the bank's marketing campaign.

2. Data Description

In Figure 1, the 47 input variables and binary target variable, "Ins", indicating whether a customer purchased the variable annuity product, are essential for the marketing campaign and variable annuity purchase decision. Demographic factors like age, homeownership, income, and residence length highlight customer segments with financial stability and long-term planning mindset. Additionally, variables such as ATM usage, credit card activity, savings balance, and retirement account ownership indicate customers actively managing finances and making investments. By targeting these segments, ABC bank can optimise resource allocation and improve campaign effectiveness. Hence, focusing on customers with a higher probability of converting to variable annuity customers allows for better resource allocation and campaign outcomes.

Input variable	Description
ATM (ATM)	This variable indicates whether the customer used the ATM service (1 = yes, 0 = no). It provides information on the customer's ATM usage behaviour.
ATMAmt (ATM Withdrawal Amount)	This variable represents the amount withdrawn by the customer using the ATM. It provides insights into the customer's cash withdrawal patterns.
AcctAge (Age of Oldest Account)	This variable indicates the age of the customer's oldest account in years. It reflects the customer's long-term relationship with the bank.
Age	This variable represents the age of the customer in years. It provides demographic information about the customer.
Branch	This variable represents the branch of the bank the customer is associated with. It is a categorical variable with 16 possible values (B1 - B19).
CC (Credit Card)	This variable indicates whether the customer has a credit card account (1 = yes, 0 = no). It provides information on the customer's credit card usage behaviour.
CCBal (Credit Card Balance)	This variable represents the balance on the customer's credit card account. It provides insights into the customer's credit card debt.
CCPurc (Credit Card Purchases)	This variable represents the number of credit card purchases made by the customer. It reflects the customer's credit card usage behaviour.
CD (Certificate of Deposit)	This variable indicates whether the customer has a certificate of deposit (CD) account (1 = yes, 0 = no). It provides information on the customer's investment preferences.
CDBal (CD Balance)	This variable represents the balance on the customer's certificate of deposit account. It provides insights into the customer's investment in CDs.
CRScore (Credit Score)	This variable represents the customer's credit score. It provides information on the customer's creditworthiness.
CashBk (Number Cash Back)	This variable represents the number of times the customer received cash back. It reflects the customer's cash usage behaviour.
Checks (Number of Checks)	This variable represents the number of checks used by the customer. It provides insights into the customer's check-writing behaviour.
DDA (Checking Account)	This variable indicates whether the customer has a checking account (1 = yes, 0 = no). It provides information on the customer's primary banking account.
DDABal (Checking Balance)	This variable represents the balance on the customer's checking account. It provides insights into the customer's financial liquidity.
Dep (Checking Deposits)	This variable represents the number of deposits made into the customer's checking account. It reflects the customer's deposit behaviour.
DepAmt (Amount Deposited)	This variable represents the amount deposited by the customer. It provides insights into the customer's deposit patterns.
DirDep (Direct Deposit)	This variable indicates whether the customer has direct deposit set up (1 = yes, 0 = no). It reflects the customer's income source and banking convenience.
HMOwn (Owns Home)	This variable indicates whether the customer owns a home (1 = yes, 0 = no). It provides information on the customer's homeownership status.
HMVal (Home Value)	This variable represents the value of the customer's home in thousands of dollars. It provides insights into the customer's housing assets.
ILS (Instalment Loan)	This variable indicates whether the customer has an instalment loan (1 = yes, 0 = no). It provides information on the customer's borrowing behaviour.
ILSBal (Loan Balance)	This variable represents the balance on the customer's instalment loan. It provides insights into the customer's outstanding loan amount.
IRABal (IRA Balance)	This variable represents the balance on the customer's retirement account. It provides insights into the customer's retirement savings.
InArea (Local Address)	This variable indicates whether the customer has a local address (1 = yes, 0 = no). It provides information on the customer's geographical proximity to the bank.
Income	This variable represents the customer's income in thousands of dollars. It provides demographic information about the customer's financial position.
Ins (Purchase Variable Annuity Account)	This is the binary target variable. It indicates whether the customer purchased the variable annuity product (1 = yes, 0 = no). The objective of the predictive model is to accurately predict this variable based on the other input variables.
Inv (Investment)	This variable indicates whether the customer has an investment account (1 = yes, 0 = no). It provides information on the customer's investment behaviour.
InvBal (Investment Balance)	This variable represents the balance amount in the customer's investment account.

LOC (Line of Credit)	This variable indicates whether the customer has a line of credit (1 = yes, 0 = no). It provides information on whether the customer has access to a line of credit.
LOCBal (Line of Credit Balance)	This variable represents the balance on the customer's line of credit account. It provides insights into the customer's utilisation of the line of credit.
LORes (Length of Residence)	This variable represents the length of time the customer has resided at their current residence in years. It provides information on the stability of the customer's residential address.
MM (Money Market)	This variable indicates whether the customer has a money market account (1 = yes, 0 = no). It provides information on the customer's investment preferences.
MMBal (Money Market Balance)	This variable represents the balance on the customer's money market account. It provides insights into the customer's investment in money market instruments.
MMCred (Money Market Credits)	This variable represents the number of credits or deposits made into the customer's money market account. It reflects the customer's deposit behaviour with respect to the money market account.
MTG (Mortgage)	This variable indicates whether the customer has a mortgage account (1 = yes, 0 = no). It provides information on the customer's homeownership status and mortgage obligations.
MTGBal (Mortgage Balance)	This variable represents the outstanding balance on the customer's mortgage account. It provides insights into the customer's mortgage debt.
Moved (Recent Address Change)	This variable indicates whether the customer has recently changed their address (1 = yes, 0 = no). It provides information on the customer's recent relocation.
NSF (Number Insufficient Fund)	This variable indicates the occurrence of insufficient funds in the customer's accounts (1 = yes, 0 = no). It reflects instances where the customer did not have sufficient funds to cover a transaction.
NSFAmt (Amount NSF)	This variable represents the amount of insufficient funds in the customer's accounts. It provides insights into the magnitude of the insufficient funds occurrences.
POS (Number Point of Sale)	This variable represents the number of point of sale transactions made by the customer. It reflects the customer's usage of their accounts for purchases at merchant locations.
POSAmt (Amount Point of Sale)	This variable represents the total amount spent by the customer in point of sale transactions. It provides insights into the customer's spending behaviour during point of sale transactions.
Phone (Number Telephone Banking)	This variable represents the number of times the customer used telephone banking services. It reflects the customer's utilisation of telephone banking for account management.
Res (Area Classification)	This variable represents the area classification of the customer's residential address. It is a categorical variable with three possible values: R (rural), S (suburb), and U (urban).
SDB (Safety Deposit Box)	This variable indicates whether the customer has a safety deposit box at the bank (1 = yes, 0 = no). It provides information on the customer's usage of safety deposit box services.
Sav (Savings Account)	This variable indicates whether the customer has a savings account (1 = yes, 0 = no). It provides information on the customer's savings behaviour.
SavBal (Savings Balance)	This variable represents the balance on the customer's savings account. It provides insights into the customer's savings habits.
Teller (Teller Visits)	This variable represents the number of visits made by the customer to a teller at the bank. It reflects the customer's preference for in-person banking interactions.

Figure 1: Input variables and binary target description

After going through the data exploration process, I noticed that there were 15 input variables that contained missing values which could introduce bias and affect the accuracy of this predictive model. In Figure 2, you can see there are input variables that contain missing values. For example, the “Age” variable has the highest missing value of 19.7%, followed by variables, “HMVal”, “income”, and “LORes” with 17.9% of missing values.

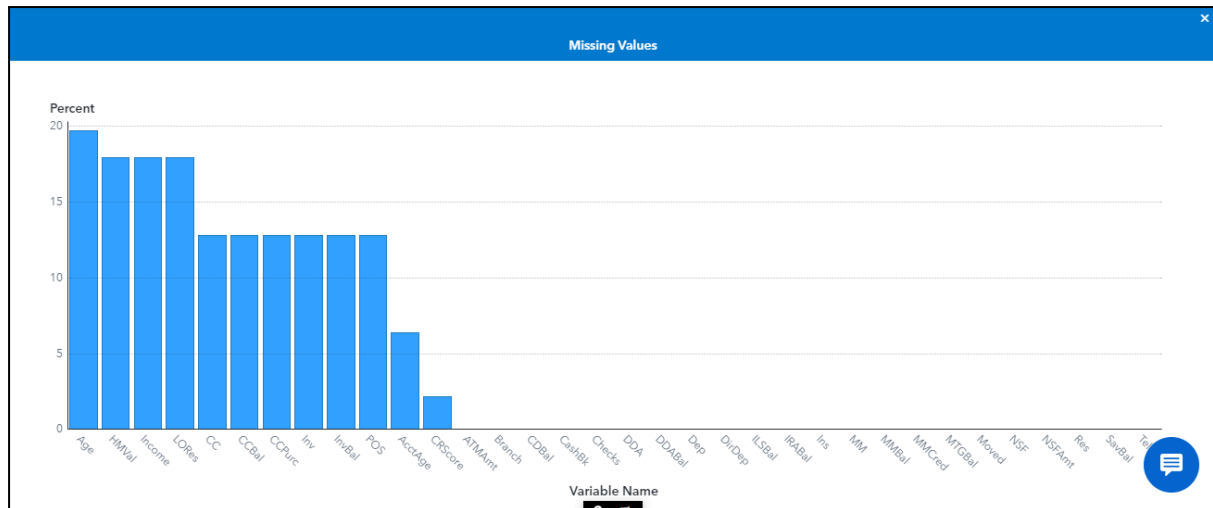


Figure 2 : The percentage of missing values for each variable obtained from data exploration

Variable Name	Label	Type	Minimum ↑	Role
InvBal	Investment Balance	Numeric	-2,214.9200	Input
CCBal	Credit Card Balance	Numeric	-2,060.5100	Input
DDABal	Checking Balance	Numeric	-774.8300	Input
LOCBal	Line of Credit Balance	Numeric	-613.0000	Input
branch_swoe		Numeric	-1.4202	Rejected
ATM	ATM	Numeric	0.0000	Input
ATMAmt	ATM Withdrawal Amount	Numeric	0.0000	Input
CashBk	Number Cash Back	Numeric	0.0000	Input

Figure 3 : Multiple input variables consisting of negative values in the minimum column

In Figure 3, there are 5 variables that have negative minimum values. However, I have decided not to remove these variables as it is logical for them to have negative values. As for the variable “branch_swoe”, it has already been rejected beforehand because it has low relevance to the objective so the negative value does not affect the model.

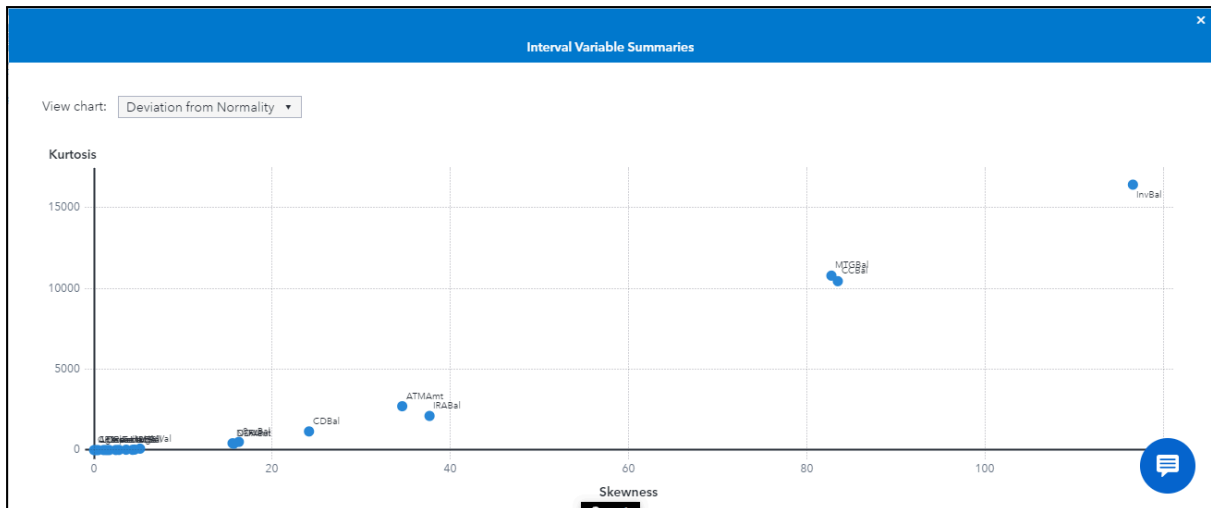


Figure 4 : Interval Variable Summaries obtained from Data exploration results.

Figure 4 shows the skewness and kurtosis from the Interval Variable Summaries. I can see that there are some variables that have high kurtosis and high skewness as they are positioned at the upper right area. This indicates that the data has a high tendency for outliers.

3. Methodology

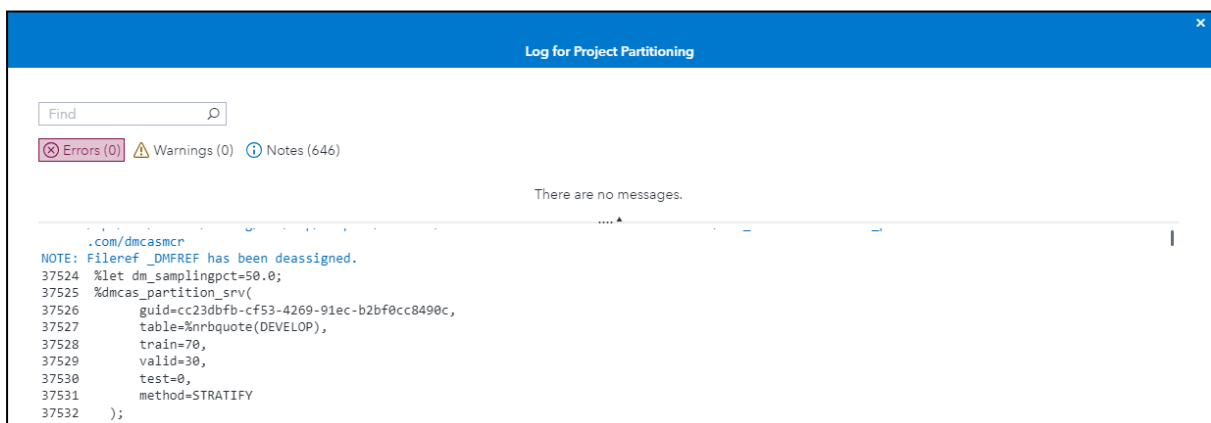


Figure 5: Data separation

In Figure 5, for the data separation, I used the stratify method and I have allocated 70% for the training data set so the model has more data to learn from, potentially leading to better performance. The validation set, comprising 30%, is used to evaluate the model's performance on unseen data to allow for fine-tuning and optimization. The variables in Figure 3 showing negative minimum values are added with a lower limit of 0.0 in the metadata but this does not change the value of the variables. However, I have decided not to

add a replacement node to the pipeline as variables with negative minimum values are logical.

Variable Na...	Minimum	Maximum	Mean	Standard D...	Skewness ↓	Kurtosis	Relative Va...	Mean plus ...	Mean minu...
InvBal	-2,214.9200	8,323,796.0200	1,599.1706	56,940.1267	116.6110	16,407.4461	35.6060	115,479.4240	-112,281.0827
CCBal	-2,060.5100	10,641,354.7800	9,586.5525	81,383.9495	83.4902	10,441.0944	8.4894	172,354.4514	-153,181.3464
MTGBal	0	10,887,573.2800	8,081.7360	79,794.1155	82.7724	10,782.4726	9.8734	167,669.9670	-151,506.4950
IRABal	0	596,497.6000	617.5705	7,516.5648	37.6669	2,111.7820	12.1712	15,650.7001	-14,415.5592
ATMAmt	0	427,731.2600	1,235.4147	4,462.2819	34.6011	2,712.1869	3.6120	10,159.9785	-7,689.1491
CDBal	0	1,053,900	2,530.7091	14,811.7810	24.1407	1,154.4064	5.8528	32,154.2711	-27,092.8528
SavBal	0	700,026.9400	3,170.6037	13,397.1469	16.2641	507.0017	4.2254	29,964.8975	-23,623.6901
DDABal	-774.8300	278,093.8300	2,170.0167	7,282.7270	15.6940	390.3279	3.3561	16,735.4707	-12,395.4373
NSFAmt	0	666.8500	2.2905	14.3637	15.5582	423.1459	6.2709	31.0179	-26.4368
HMVAl	67	754	110.9121	21.9499	5.1730	89.6587	0.1979	154.8120	67.0121
POS	0	54	1.0757	2.8887	4.5579	31.3488	2.6854	6.8530	-4.7723

Figure 6 : Skewness and Kurtosis table obtained from data exploration

In Figure 6, there are variables with extremely high skewness and kurtosis which means transformation is needed. Most of the variables are balances and amounts. In the metadata, the variables in Figure 6 are selected and transformed into log to normalise the right-skewed distribution displayed in Figure 7.

Variable Name	Label	Type	Role	Level
☒ MTGBal	Mortgage Balance	Numeric	Input	Interval
☐ NSF	Number Insufficient Fund	Numeric	Input	Binary
☒ NSFAMt	Amount NSF	Numeric	Input	Interval
☐ off		Numeric	Rejected	Unary
☐ Phone	Number Telephone Banking	Numeric	Input	Interval
☒ POS	Number Point of Sale	Numeric	Input	Interval
☐ POSAMt	Amount Point of Sale	Numeric	Input	Interval
☐ Res	Area Classification	Character	Input	Nominal
☐ Sav	Saving Account	Numeric	Input	Binary
☒ SavBal	Saving Balance	Numeric	Input	Interval

Multiple Variables

Role:

Level:

Order:

Transform:

Impute:

Figure 7: Transformation of variables to log

After adding the transformation node, I kept all the default settings so all the input variables will transform based on the metadata setting, which is the log method. In Figure 8, SAS Viya has created the formulas for the new transformed variables. Some variables are added with +1 in the formula to avoid any variables being undefined and other positive values have changed to a smaller value. Hence, right-skewed distributions will become symmetric.

Transformed Variables Summary						
Transformed Variable	Method	Input Variable	Formula	Variable Level	Type	Variable Label
LOG_ATMAmt	LOG	ATMAmt	$\log('ATMAmt' \cdot n + 1)$	INTERVAL	N	Transformed ATM Withdrawal Amount
LOG_CCBal	LOG	CCBal	$\log('CCBal' \cdot n + 2061.51)$	INTERVAL	N	Transformed Credit Card Balance
LOG_CDBal	LOG	CDBal	$\log('CDBal' \cdot n + 1)$	INTERVAL	N	Transformed CD Balance
LOG_DDABal	LOG	DDABal	$\log('DDABal' \cdot n + 400.53)$	INTERVAL	N	Transformed Checking Balance
LOG_HMVal	LOG	HMVal	$\log('HMVal' \cdot n + 1)$	INTERVAL	N	Transformed Home Value
LOG_ILSBal	LOG	ILSBal	$\log('ILSBal' \cdot n + 1)$	INTERVAL	N	Transformed Loan Balance
LOG_InvBal	LOG	InvBal	$\log('InvBal' \cdot n + 2215.92)$	INTERVAL	N	Transformed Investment Balance

Figure 8 : New log transformation variables

As mentioned previously, there are multiple inputs that have missing values so an imputation node is added to replace the missing values. I have created another pipeline to test whether the default interval input (mean) or the changed interval input (median) will produce a better result. However, there were no significant differences in the misclassification rate so I proceeded with using mean. Also, I did not add a text mining node as there were no text variables. For feature selection, as this data set has many variables, I added the variable clustering node to reduce variable redundancy. In Figure 9, I have remained all default settings except I have ticked the “include class inputs” so the class variables are included in the clustering process and I unticked the “export class level indicators” because there is no need for me to check on the variable levels.

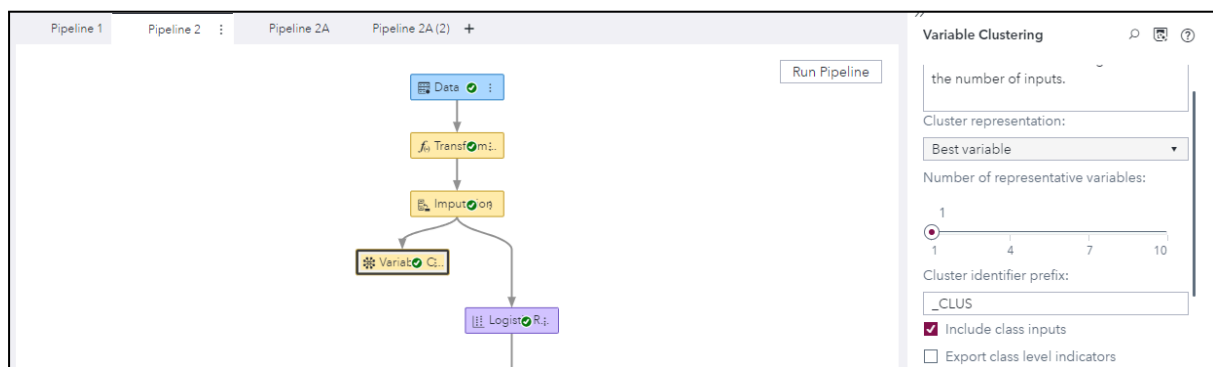


Figure 9 : Variable clustering settings

Cluster ID	Variable	Variable Label	Variable Selected	Principal Component 1
_CLUS1	ATM	ATM	YES	-0.6475
_CLUS1	DDA	Checking Account	NO	-0.4188
_CLUS1	LOG_ATMAmt	Transformed ATM Withdrawal Amount	NO	0.6367
_CLUS2	LOG_CDBal	Transformed CD Balance	YES	0.7071
_CLUS2	CD	Certificate of Deposit	NO	-0.7071
_CLUS3	MM	Money Market	YES	-0.7071
_CLUS3	LOG_MMBal	Transformed Money Market Balance	NO	0.7071
_CLUS4	LOG_SavBal	Transformed Saving Balance	YES	0.7071
_CLUS4	Sav	Saving Account	NO	-0.7071

Figure 10 : Results of Clustered variables table

In Figure 10, the ATM variable has been selected as the best variable to represent the first cluster whereas the DDA and LOG_ATMAmt variable is redundant for the first cluster. The same logic applies to Cluster 2, 3 and 4.

Next, a variable selection node is added to reduce the number of inputs by filtering variables that are unrelated to the target variable. Firstly, I separated the variable selection from the variable clustering node and I got a misclassification rate of 0.2647 but initially I got 0.2636 after imputation. So I thought of adding variable selection after the variable clustering node, however, the misclassification rate increased to 0.2863. Hence, I decided to experiment with the settings for variable selection under linear regression selection (refer Figure 11).

Figure 11: Variable selection settings

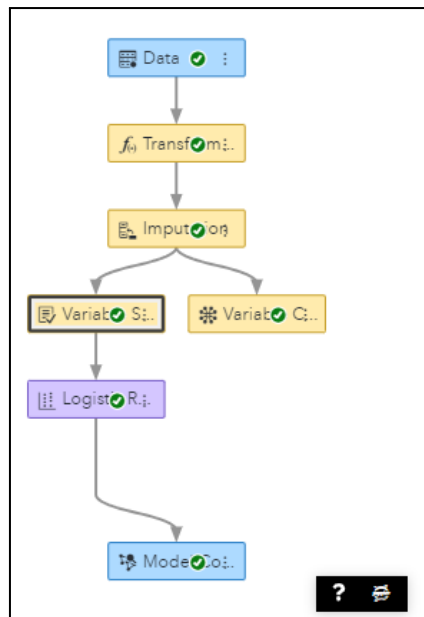


Figure 12 : The final model for selection process

Model Comparison					Download
Champion	Name	Algorithm Name	KS (Youden)	Misclassification Rate	
	Logistic Regression (1)	Logistic Regression	0.4437	0.2637	

Figure 13 : Results of logistic regression - forward

In Figure 11, I have chosen the unsupervised selection to reduce the redundant variables, fast-supervised selection to observe the relevancy of the input. For linear regression selection, the method I chose was forward and the criterion, Validation ASE. This method produced the best results compared to stepwise and backward selection with a misclassification rate of 0.2637 (refer Figure 13). Therefore, the final model for the selection process is finalised (refer Figure 12).

4. Model development

For the model development process, I wanted to experiment with multiple pipelines to see which model would produce the best results.

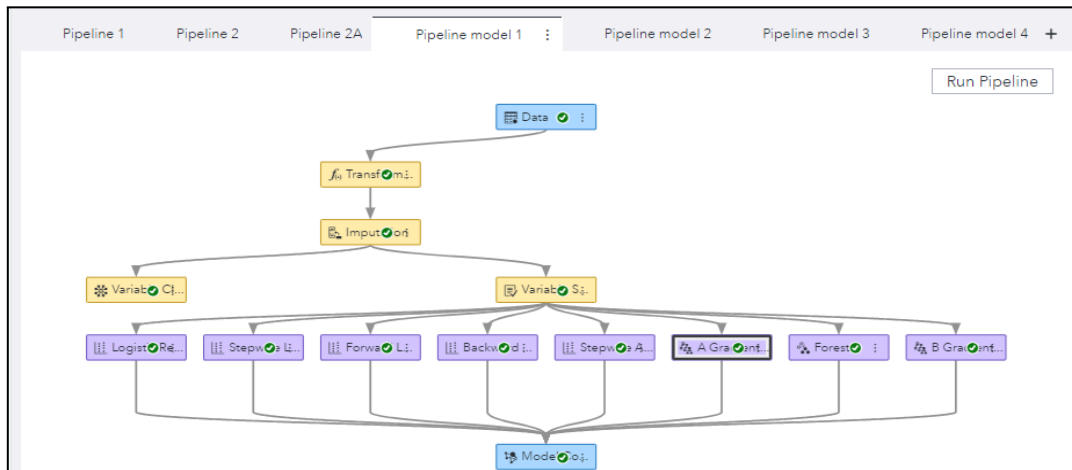


Figure 14 : First pipeline model

Champion	Name	Algorithm Name	KS (Youden)	Misclassification Rate
	Gradient Boosting	Gradient Boosting	0.4676	0.2529
	B Gradient Boosting	Gradient Boosting	0.4610	0.2477
	Forest	Forest	0.4590	0.2569
	Stepwise Logistic Regression	Logistic Regression	0.4403	0.2647
	Stepwise A Logistic Regression	Logistic Regression	0.4403	0.2647
	Forward Logistic Regression	Logistic Regression	0.4403	0.2647
	Backward Logistic Regression	Logistic Regression	0.4403	0.2647
	Logistic Regression (1)	Logistic Regression	0.4386	0.2637

Figure 15 : Model comparison results - first pipeline

The first pipeline I created a forward, backward and stepwise regression models for the selection method under settings. After running the model comparison node, I realised all models produced the same result so I decided to change the settings such as significance level, model selection criterion and a few more functions but there were still no changes among their misclassification rate. Hence, I added 2 more gradient boosting nodes with different parameters and a forest node. Figure 14 shows the gradient boosting with the default settings has the lowest misclassification rate.

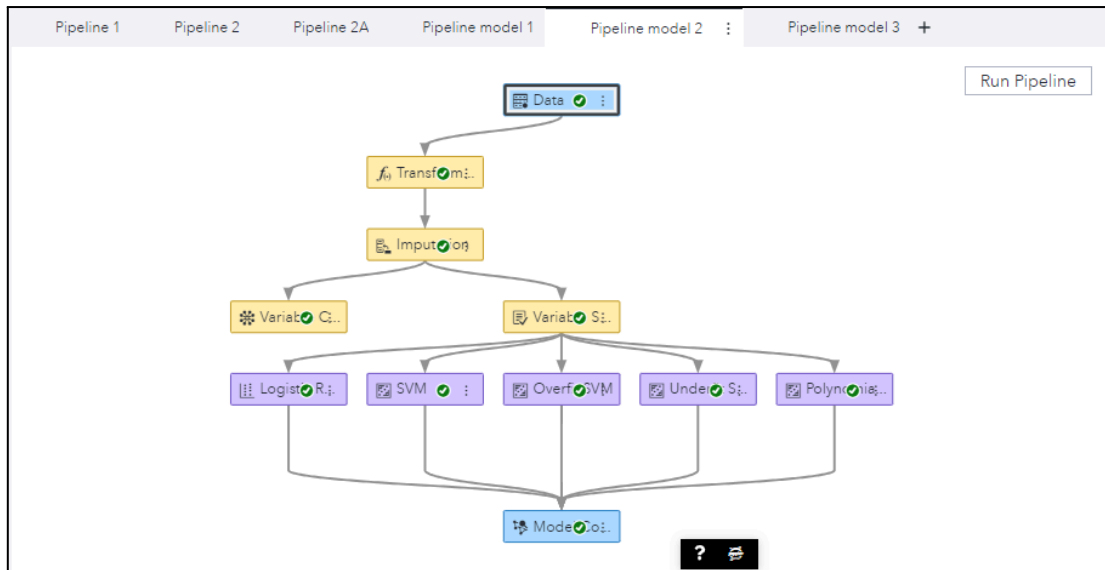


Figure 16 : Second pipeline model

In Figure 16, I wanted to compare the performance between SVM models so I created an Overfitting SVM, an Underfitting SVM and a Polynomial SVM. I experimented with multiple different settings including changing the Kernel, Penalty value, Tolerance value and number of maximum iterations. However, after running model comparison, the SVM node with a changed tolerance value of 0.6 was the champion model for the second pipeline.

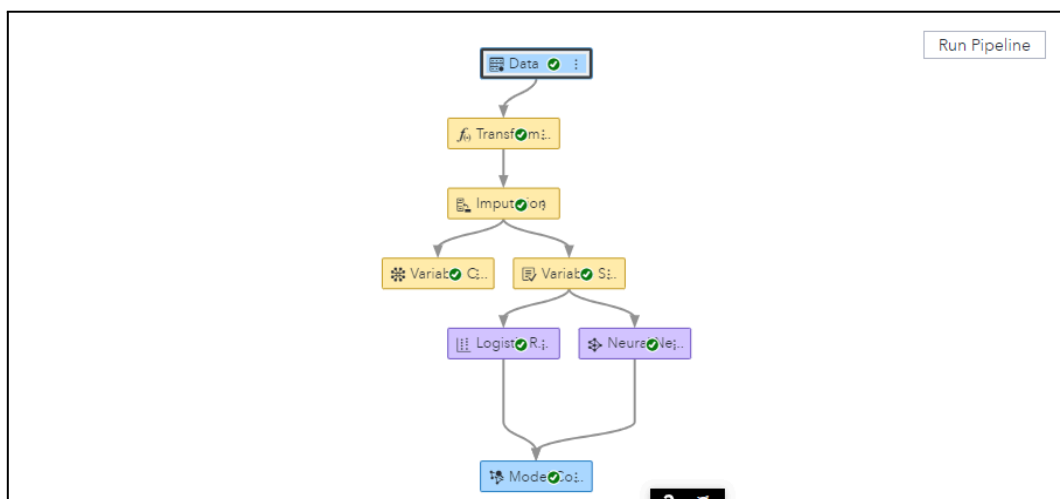


Figure 17 : Third pipeline model

In Figure 17, I tested a neural network node connected to the variable selection node. I decided to change the default settings under the neural network, common optimization section, because the default model produced very poor performance. I reduced the values for L1 and L2 weight decay to 0.01 and 0.0001 respectively, so it will control the weight for

variables that are not significant. Ultimately, the results had improved drastically but still not the best model.

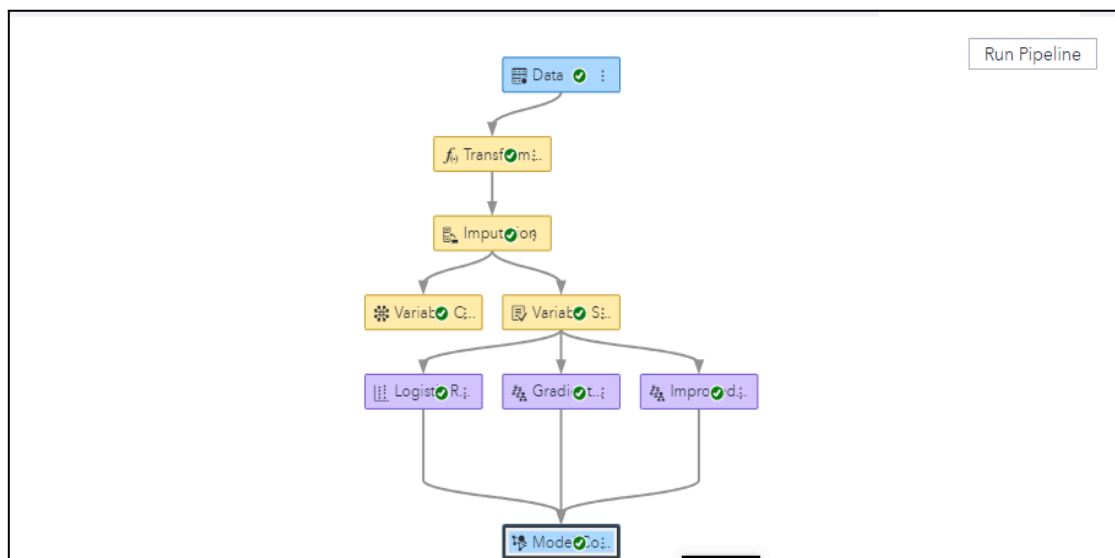


Figure 18 : Fourth pipeline model

After producing these 3 pipelines, I was still not satisfied with the result so I created a fourth pipeline (refer Figure 18). I wanted to compare the gradient boosting separated from the logistic regression comparison pipeline. Hence, I kept the first gradient boosting node with default settings (refer Figure 19) and changed the parameters for L1 regularisation of the second gradient boosting node to 0.8 (refer Figure 20). I tried to change the number of trees as well but it would reduce the KS (Youden) value so it is not so relevant to this model.

Gradient Boosting

Fits a gradient boosting model, which builds a sequential series of decision trees.

Basic Options

Number of trees: 100

Learning rate: 0.1

Subsample rate: 0.5

L1 regularization: 0

L2 regularization: 1

Interval target distribution: Normal

Figure 19 : Default Setting

Improved Gradient Boosting

Fits a gradient boosting model, which builds a sequential series of decision trees.

Basic Options

Number of trees: 100

Learning rate: 0.1

Subsample rate: 0.5

L1 regularization: 0.8

L2 regularization: 1

Interval target distribution: Normal

Figure 20: Hyperparameter Tuning

Model Comparison				
Champion	Name	Algorithm Name	KS (Youden)	Misclassification Rate
☒	Improved Gradient Boosting	Gradient Boosting	0.4716	0.2464
☐	Gradient Boosting	Gradient Boosting	0.4690	0.2475
☐	Logistic Regression (1)	Logistic Regression	0.4437	0.2637

Figure 21 : Model comparison results - pipeline four

In Figure 21, the Improved Gradient boosting node has the lowest misclassification rate after doing hyperparameter tuning. I thought increasing the L1 regularisation parameter too much can potentially lead to underfitting because the model could be overly simplified and fails to capture important patterns in the data. However, after tuning the L1 parameters, it improved the performance metrics by reducing the misclassification rate. This indicates the model is benefiting from feature selection and finding a better balance between bias and variance for the problem.

Data Pipelines Pipeline Comparison Insights							
Filter		Data: Validate		Compare			
☐	Champion ↓	Name	Algorithm Name	Pipeline Name	Misclassification Rate	KS (Youden)	Sum of Frequencies
☒	☒	Improved Gradient ...	Gradient Boosting	Pipeline model 4	0.246	0.472	9,680
☐	☐	A Gradient Boosting	Gradient Boosting	Pipeline model 1	0.253	0.468	9,680
☐	☐	SVM	SVM	Pipeline model 2	0.258	0.448	9,680
☐	☐	Logistic Regression	Logistic Regression	Pipeline 2	0.264	0.444	9,680
☐	☐	Logistic Regression (1)	Logistic Regression	Pipeline 2A	0.264	0.444	9,680
☐	☐	Logistic Regression (1)	Logistic Regression	Pipeline model 3	0.264	0.444	9,680

Figure 22 : Pipeline comparison results

Then, I checked the pipeline comparison and it is evident in Figure 22 that the champion model identified for this project is the Improved Gradient Boosting from pipeline 4. It has the lowest misclassification rate of 0.246 compared to the other models, demonstrating its effectiveness in accurately predicting the binary outcome. The binary nature of the target variable makes this misclassification rate particularly relevant and meaningful as minimising the misclassification rate can identify potential customers who are more likely to purchase the variable annuity product accurately.

5. Discussion

With the champion model, ABC bank can use the predictions to focus their marketing efforts on customers who are more likely to respond positively to the campaign. By achieving a low misclassification rate, the champion model is successfully fulfilling the objective of accurately identifying customers who are more likely to purchase the product.

As for the champion model's alignment with business goals, the model predictions can be integrated into ABC bank's decision-making processes for marketing, customer segmentation, and resource allocation.. The model can be regularly updated and retrained with new data to adapt to changing customer behaviours and market dynamics. Besides that, the model's accuracy in identifying potential purchasers reduces the risk of targeting the wrong customers or investing in ineffective marketing strategies. Thus, leading to cost reductions by avoiding unnecessary expenditures on unresponsive customer segments.

6. Conclusion

In conclusion, the predictive modelling process to address ABC bank's target marketing campaign for the variable annuity product has significant business implications. The champion model, Improved Gradient Boosting from pipeline 4, accurately identified potential purchasers with a low misclassification rate of 0.246. Through a meticulous methodology, analysing binary target and input variables including demographic information and product usage patterns allows the model to effectively address the business problem. Also, integrating the model into business processes will result in optimised marketing strategies, resource allocation, and personalised communication to potential purchasers. Moreover, regular updates and retraining of the model ensures adaptability to changing customer behaviours and market dynamics. The process emphasises the importance of data-driven insights in driving successful target marketing campaigns and achieving desired business objectives.